

Experiment 01

Aim:

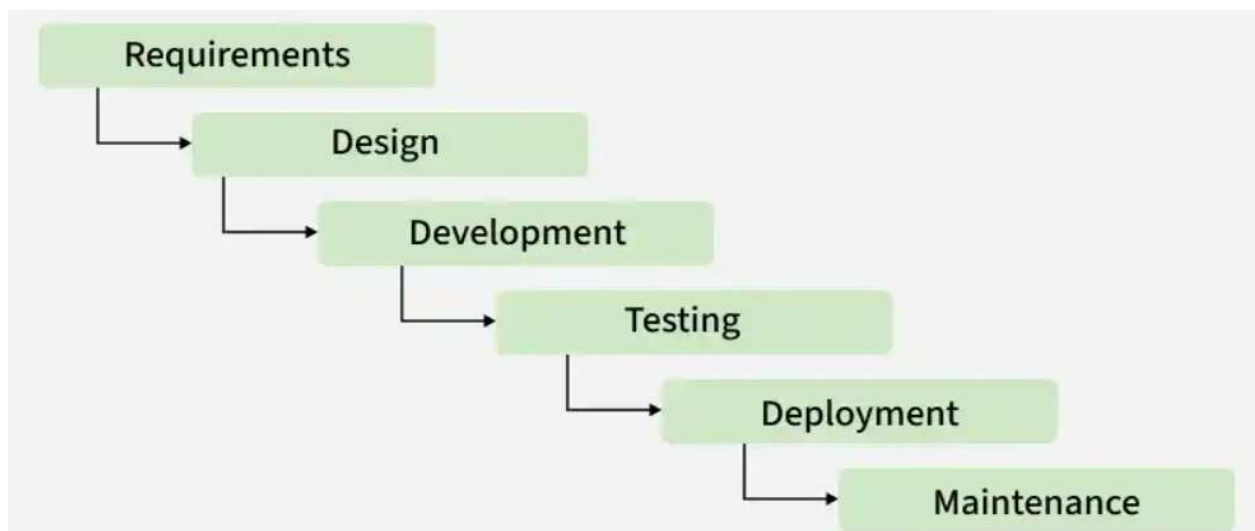
To study the **Software Process Model** and Detailed Software Project (DSP) for **Hostel Management System**.

Model Used: Waterfall Model

Problem Statement:

The Hostel Management System is designed to automate hostel activities such as student registration, room allocation, fee management, attendance, complaint handling, and report generation. The existing manual system is time-consuming, error-prone, and difficult to maintain. The proposed system improves efficiency, accuracy, and security by maintaining all hostel records digitally.

Diagram:



Why the Waterfall Model is Chosen

- Hostel management requirements are clear and well-defined.
- The system has fixed functionalities such as student registration, room allocation, fee management, and report generation.
- Documentation is maintained at every stage.
- It is simple to understand and implement.
- Suitable for academic and small-scale software projects.
- Easier to manage due to its sequential development approach.

Phases:

1. Requirement Analysis

- Gather and analyze all project requirements.
- Define functional and non-functional requirements.

2. Design

- Design the overall system architecture.
- Prepare database design, user interface (UI), and module design.
- Define how each module will interact.

3. Development

- Develop the Hostel Management System by writing the program code.
- Create modules such as Student Management, Room Management, Fee Management, Attendance Management, and Complaint Management.

4. Testing

- Test the software to identify and fix errors.
- Verify that all modules work correctly according to the requirements.

5. Deployment

- Install and configure the Hostel Management System for users.
- Make the software available to the hostel administrator and students.
- Provide user training if required.

6. Maintenance

- Fix bugs and errors after deployment.
- Update the system based on user needs.
- Improve performance and add new features whenever required.

Outcome:

The study of the **Waterfall Software Process Model** was successfully completed, and a **Detailed Software Project (DSP)** for the **Hostel Management System** was prepared. The proposed system effectively automates hostel operations, reduces manual work, improves data accuracy, and provides secure and efficient management of hostel records.